

AI-Based Fake Identity & Document Screening

Research & Model-Gap Report

SIH26188 • Evidence review for selecting a research problem and model direction

Purpose: This report does not jump directly to a new model. It studies existing approaches, datasets and reported limitations to identify a defensible research gap that can be experimentally tested by a student team.

Question	Current evidence
What do existing systems already solve?	Document analysis, forgery/PAD detection, face verification and anti-spoofing are active research areas.
Where do they struggle?	Unseen attacks, cross-domain generalization, realistic capture conditions, data scarcity and privacy constraints repeatedly appear.
Most promising gap	Robust detection of unseen/realistic identity-document attacks and generalization across capture conditions.
Practical project strategy	Reproduce a strong baseline first, deliberately test it on unseen attacks/domain shifts, then add one focused improvement.

Executive conclusion

The literature does not suggest that the main problem is simply building another classifier. A recurring weakness is **generalization**: models can perform strongly on the attacks and datasets they were trained/evaluated on, while performance becomes harder to guarantee when the attack type, document type, camera, illumination, compression or domain changes. The 2025 identity-card systematic review explicitly identifies poor generalization and a gap between controlled/private datasets and public/realistic evaluation. FakeIDet2 also makes unseen-attack evaluation a central part of its benchmark. This makes unseen/realistic attack robustness a strong candidate research direction.

1. Research Objective

The team's current research objective should be framed as:

“Can an existing AI-based identity-document forgery detector be made more robust to previously unseen manipulations and real-world capture/domain changes?”

This is intentionally a research question rather than a fixed architecture. The architecture should be selected only after the baseline failures are measured.

Why this direction?

- The 2025 systematic review reports poor generalization across document types and new attack methods.
- FakeIDet2 explicitly evaluates realistic fake IDs under unseen attack types and provides a reproducible benchmark.
- MIDV-2020 and IDNet show that dataset diversity and privacy-preserving data availability are central research issues.
- Face anti-spoofing literature independently shows the same open-set/cross-domain challenge.

This convergence across document PAD and face PAD is useful: it means the proposed gap is not based on one paper alone.

2. Research Paper Matrix

The table below is designed for the team's literature-review workflow. "Limitation" means a limitation or remaining challenge supported by the paper/source description; it is not a claim that the authors failed to solve every aspect of the issue.

Paper	Model / method	Main limitation / gap to test
Identity Card Presentation Attack Detection: A Systematic Review (2025)	Reviews CNNs, forensic micro-artifact methods and newer foundation-model approaches	Poor generalization, limited data, a "Reality Gap" between private/large datasets and public/mock datasets, and a "Synthetic Utility Gap."
Privacy-aware detection of fake identity documents: methodology, benchmark, and improved algorithms (FakeIDet2) (2025/2026)	Patch Embedding Extractor + Patch Embedding Fusion	The paper improves unseen-attack performance, but the existence of a dedicated unseen-attack benchmark itself shows that attack generalization remains a central evaluation challenge.
EdgeDoc: Hybrid CNN-Transformer Model for Accurate Forgery Detection and Localization in ID Documents (2025)	Lightweight convolutional transformer + NoisePrint auxiliary features; fusion with TruFor explored.	Strong benchmark performance does not remove the broader cross-dataset/generalization problem; the authors also explore fusion to improve generalization.
Improving Presentation Attack Detection for ID Cards on Remote Verification Systems (2023)	Two-stage end-to-end approach based on MobileNetV2.	The approach is trained/evaluated around defined attack species and a particular data environment, motivating tests beyond those known conditions.
Open Set Face Anti-Spoofing in Unseen Attacks (2021)	End-to-end open-set face anti-spoofing approach.	This is face PAD rather than document PAD; the transfer of the open-set idea to identity-document attacks is an opportunity to investigate.
Anomaly Detection-Based Unknown Face Presentation Attack Detection (2020)	Deep-learning anomaly detector with pseudo-negative class and pairwise confusion loss.	Focused on face PAD; document-specific unknown forgery detection remains a separate challenge.
MIDV-2020: A Comprehensive Benchmark Dataset for Identity Document Analysis (2021/2022)	Provides baselines for document location, identification, text recognition and face detection.	Uses mock/synthetic document content; the broader literature still faces the gap between public datasets and realistic private identity documents.
IDNet: A Novel Dataset for Identity Document Analysis and Fraud Detection (2024)	Dataset and use-case benchmark rather than one detector.	Synthetic data is valuable for scale and privacy, but must be tested for transfer to real capture conditions.
Identity Documents Authentication based on Forgery Detection of Guilloche Pattern (2022)	CNN feature extraction + similarity measurement for Guilloche pattern conformity.	A feature-specific detector can be vulnerable when the attack does not affect the targeted security pattern or when document designs differ.
Deep Transfer Learning on the Aggregated Dataset for Face Presentation Attack Detection (2022)	Deep transfer learning.	Face PAD remains affected by domain shifts and dataset differences; this supports cross-domain evaluation as a research criterion.

3. What Existing Models Already Solve

Document presentation attacks

MobileNetV2-based systems can detect defined attack classes such as printed, display, composite, plastic and synthetic ID cards.

Patch-level fake-ID detection

FakeIDet2 moves beyond a simple whole-image classifier by extracting and fusing patch embeddings, while also addressing privacy-aware data collection.

Forgery localization

EdgeDoc combines image content with forensic NoisePrint features and targets both detection and localization of manipulated regions.

Open-set face anti-spoofing

Face PAD research has explicitly formulated the problem of detecting attack types not present in training.

Anomaly-based detection

Anomaly detection can learn from genuine/non-attacked samples and use deviations as evidence of unknown attacks.

Dataset engineering

MIDV-2020 and IDNet demonstrate that dataset construction is itself a major research contribution because identity documents are sensitive and difficult to collect.

4. The Repeated Failure Pattern

Pattern observed across the literature:

Known attack → controlled dataset → strong performance → new attack/domain/capture condition → uncertain generalization.

For our team, this should become an experimental hypothesis rather than an assumption.

Hypothesis H1: A strong existing detector will lose measurable performance when tested on attack types, document/capture conditions, or data sources that are not represented in its training distribution.

Hypothesis H2: Adding a mechanism designed to learn more general forensic evidence—such as anomaly/open-set features, multi-scale forensic features, domain-invariant representations, or evidence fusion—can reduce that performance drop.

5. Recommended Research Direction

Primary direction: Robust / open-set fake identity-document detection under unseen attacks and real-world domain shifts.

Do not start by inventing a neural architecture. Start with a baseline-and-gap experiment.

Recommended experimental pipeline

1. Select baseline

Choose one reproducible strong model, preferably FakeIDet2 or EdgeDoc; keep a lighter MobileNetV2-style baseline for comparison.

2. Reproduce baseline

Train/evaluate under the paper's intended protocol as closely as possible.

3. Create a deliberate unseen-attack split

Train on attack families A/B/C and hold out attack family D. Do not leak attack type or near-duplicate images across train/test.

4. Add domain shift

Change capture conditions: phone model, illumination, perspective, blur, JPEG compression, screen/print capture or document template.

5. Measure the failure

Report EER/HTER/AUC/F1 as appropriate, plus confusion matrices and per-attack results.

6. Improve one weakness

Try one focused improvement: open-set/anomaly objective, forensic feature branch, domain generalization, self-supervised pretraining, or evidence fusion.

7. Compare fairly

Compare baseline vs improved model on the same unseen-attack and cross-domain protocols.

8. Produce an explainable result

Show risk score + suspicious region/feature evidence rather than only a binary fake/genuine label.

6. Candidate Novelty Options

Direction	What changes	Why it is interesting	Difficulty
Open-set document PAD	Train to flag unknown manipulations rather than only known classes.	Directly targets unseen attacks.	High
Synthetic-to-real generalization	Train with synthetic/public data and optimize transfer to real mobile captures.	Addresses privacy/data scarcity + reality gap.	High
Domain-generalized document detector	Learn features stable across phone, lighting, compression and document sources.	Practical remote-KYC problem.	Medium-High
Forensic + semantic fusion	Combine forensic pixel cues with OCR/layout/field consistency.	Could catch cases where one signal looks normal.	Medium

Direction	What changes	Why it is interesting	Difficulty
Explainable risk engine	Fuse multiple model scores and localize suspicious regions.	Strong SIH demonstration value; less novel alone.	Medium

Current recommendation: Investigate open-set/unseen-attack robustness first. If the team cannot obtain a suitable document-forgery dataset/protocol, fall back to cross-domain generalization using public identity-document datasets.

7. Datasets & Reproducibility

The dataset problem is not a side issue. It may be the central reason why identity-document fraud models are difficult to compare fairly.

Dataset	Scale / content	Use in our study	Access
FakeIDet2-db	>900K real/fake ID patches from 2,000 ID images; print/screen/composite physical attacks; varied smartphone/capture conditions.	Primary candidate for fake-ID detection and unseen-attack evaluation.	GitHub / paper
MIDV-2020	1,000 video clips + 2,000 scans + 1,000 photos; 72,409 annotated images in the paper description.	Document analysis, OCR/layout/capture robustness; mock identity documents.	Dataset page
IDNet	837,060 synthetic identity-document images across 20 types from 10 U.S. states and 10 European countries.	Synthetic-to-real and cross-template experiments.	arXiv paper
Face PAD datasets	Replay-Attack, OULU-NPU, Rose-Youtu, Spoof in Wild, etc.	Only if face/liveness becomes a secondary research track.	See anomaly-PAD paper

Important: some datasets are large or require an access request/licence. Do not assume a dataset can be downloaded immediately just because the paper is free.

8. What We Should NOT Do Yet

- Do not commit to CNN vs Transformer before establishing the failure mode.
- Do not train only on random train/test splits if the goal is unseen-attack generalization.
- Do not report only overall accuracy; use attack-wise and cross-domain metrics.
- Do not claim novelty just because a different backbone is used.
- Do not build OCR + face matching + liveness + behaviour monitoring all at once; that creates a broad demo but a weak research question.

9. Paper Download / Access List

The following links were checked during preparation of this report. “Open” means a freely accessible paper/preprint or official publication page; dataset access may have separate conditions.

1. Identity Card Presentation Attack Detection: A Systematic Review (2025)

Type: Systematic review

Why read: Best paper for defining the research gap.

Access: <https://arxiv.org/abs/2511.06056>

2. Privacy-aware detection of fake identity documents: methodology, benchmark, and improved algorithms (FakeIDet2) (2025/2026)

Type: Fake-ID detector + benchmark

Why read: Highest-priority baseline/dataset paper for document fraud.

Access: <https://arxiv.org/abs/2508.11716>

3. EdgeDoc: Hybrid CNN-Transformer Model for Accurate Forgery Detection and Localization in ID Documents (2025)

Type: Forgery detection/localization model

Why read: Excellent modern architecture to reproduce as a strong baseline.

Access: https://publications.idiap.ch/attachments/reports/2025/George_Idiap-RR-08-2025.pdf

4. Improving Presentation Attack Detection for ID Cards on Remote Verification Systems (2023)

Type: ID-card PAD model

Why read: Useful lightweight baseline for student hardware and comparison.

Access: <https://arxiv.org/abs/2301.09542>

5. Open Set Face Anti-Spoofing in Unseen Attacks (2021)

Type: Open-set face PAD

Why read: Important conceptual paper for the “unknown forgery” problem.

Access: <https://dblp.org/rec/conf/mm/Dong0CLY21>

6. Anomaly Detection-Based Unknown Face Presentation Attack Detection (2020)

Type: Anomaly-detection face PAD

Why read: Strong inspiration for one-class/open-set document anomaly detection.

Access: <https://arxiv.org/abs/2007.05856>

7. MIDV-2020: A Comprehensive Benchmark Dataset for Identity Document Analysis (2021/2022)

Type: Dataset/benchmark

Why read: Useful for reproducible experiments and dataset analysis.

Access: <https://arxiv.org/abs/2107.00396>

8. IDNet: A Novel Dataset for Identity Document Analysis and Fraud Detection (2024)

Type: Large synthetic dataset

Why read: Useful for investigating synthetic-to-real generalization.

Access: <https://arxiv.org/abs/2408.01690>

9. Identity Documents Authentication based on Forgery Detection of Guilloche Pattern (2022)

Type: Document authenticity method

Why read: Shows the value—and limitation—of narrow forensic cues.

Access: <https://arxiv.org/abs/2206.10989>

10. Deep Transfer Learning on the Aggregated Dataset for Face Presentation Attack Detection (2022)

Type: Cross-dataset face PAD

Why read: Useful supporting evidence for the generalization hypothesis.

Access: <https://doi.org/10.1007/s12559-022-10037-z>

10. Suggested Reading Order for the Team

1. **Identity Card Presentation Attack Detection: A Systematic Review** — Understand the whole field and its gaps.
2. **FakeIDet2** — Understand a current fake-ID detector, benchmark and unseen-attack setup.
3. **EdgeDoc** — Study a modern forgery-localization architecture.
4. **Improving PAD for ID Cards** — Understand a lighter MobileNetV2 baseline and attack taxonomy.
5. **MIDV-2020 + IDNet** — Understand dataset limitations and synthetic-data issues.
6. **Open Set Face Anti-Spoofing + Anomaly Detection-Based Unknown PAD** — Borrow the open-set/anomaly idea from face PAD.

11. Final Research Position

Based on the reviewed evidence, the strongest candidate problem for the SIH team is not simply “detect fake documents.” The more defensible research problem is:

“Develop and evaluate a robust AI-based identity-document fraud detector that maintains useful performance when confronted with previously unseen manipulation types and real-world capture/domain changes.”

A successful project would first establish a baseline failure, then demonstrate a measurable improvement. The final contribution should therefore be expressed experimentally:

Baseline model → unseen-attack / cross-domain test → measured failure → proposed improvement → fair re-evaluation.

This structure keeps the project research-driven. If the experiments show that a different limitation is more severe—such as semantic/OCR inconsistency, localization, or multimodal evidence fusion—the team should pivot based on evidence rather than forcing the original hypothesis.

This report is a research-planning document, not a claim that the proposed improvement has already been validated. The next step is to download/read the priority papers, reproduce the strongest baseline that is feasible, and build the model-vs-failure matrix.